Séminaire de l'UQAM pour les étudiants en actuariat/statistique

Extreme Value Analysis based on Blockwise Top-Two Order Statistics

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Bundesministerium für Bildung und Forschung
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FONA Research for sustainability
CLIMXTREME Climate Change and Extreme Events

Why Extreme Value Statistics?

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The art of extrapolation beyond the limits of observation.

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- standard statistics fail $\rightsquigarrow$ necessity to **extrapolate 'into the extremes'**
- two paradigms:
  - **Block Maxima**
  - **Peak over Threshold**

Extreme Value Statistics: Block Maxima Modeling

- Take (annual/seasonal) block maxima M_n

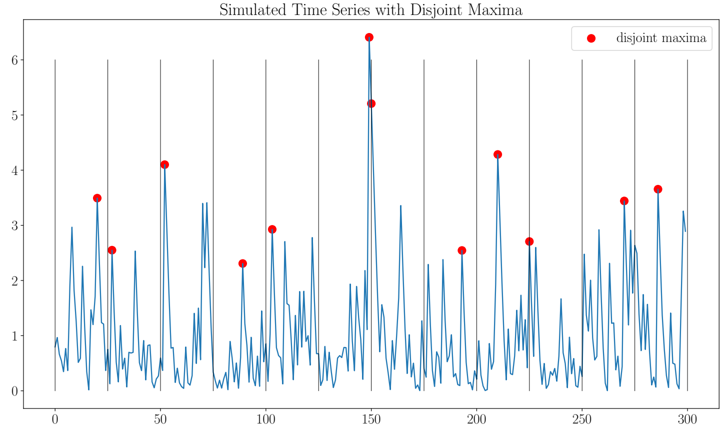

Figure 1: Sample of Disjoint Blockmaxima

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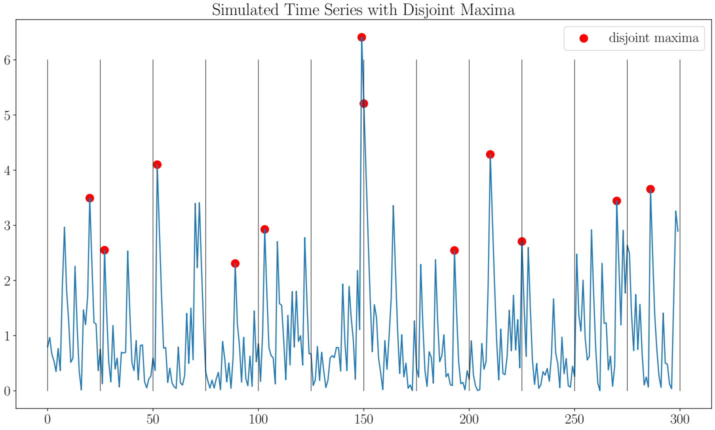
- they follow a Generalized Extreme Value Distribution (GEV)
- $M_n \sim \mathrm{GEV}(\gamma, \mu, \sigma)$



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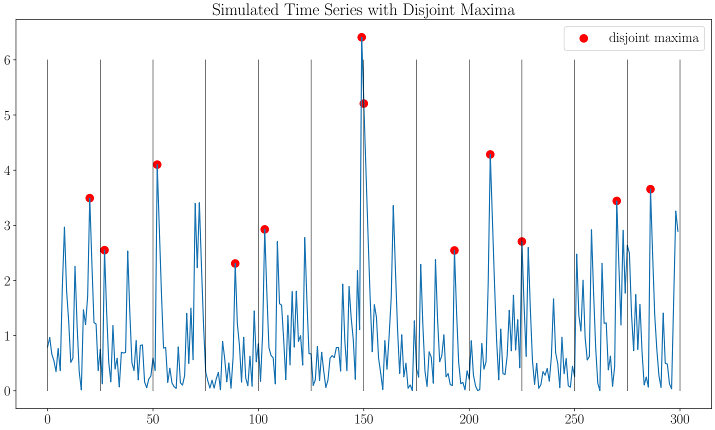
- goal: estimate (γ, μ, σ) given small number of block maxima



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Modeling Block Maxima

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Theorem (Fisher and Tippett 1928, Gnedenko 1943).

If there exist sequences $(a_r)_{r\in\mathbb{N}} \subset (0,\infty)$ and $(b_r)_{r\in\mathbb{N}} \subset \mathbb{R}$ such that $\mathbb{P}\left(\frac{M_r - b_r}{a_r} \le x\right) \xrightarrow{\mathcal{D}} G(x)$ non-degenerate, then G is one of

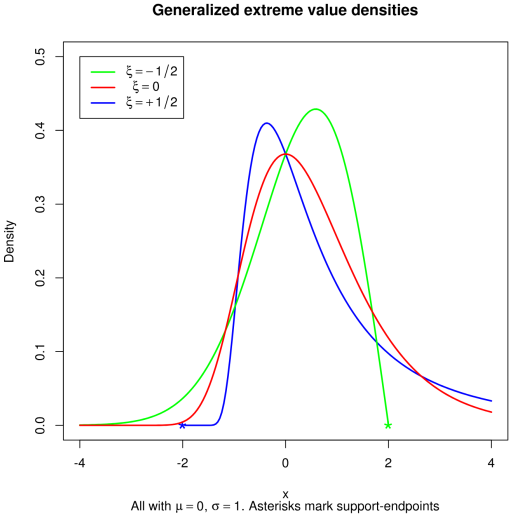
$$G(x) = \begin{cases} \exp\left(-e^{-x}\right) & x \in \mathbb{R} \quad \text{(Gumbel distribution)}, \\ \exp\left(-x^{-\alpha}\right) & \text{if } x > 0 \quad \text{(Fréchet distribution)}, \\ \exp\left(-(-x)^{\alpha}\right) & \text{if } x \le 0 \quad \text{(Weibull distribution)}. \end{cases}$$

- Three cases (Gumbel, Fréchet, Weibull) summarizable by Generalized Extreme Value (GEV) distribution

$$G_{\mathrm{GEV}(\vartheta)}(x) = \exp\left\{ -\left(1 + \xi \frac{x-\mu}{\sigma}\right)^{-1/\xi} \right\}$$

- $\vartheta = (\mu, \sigma, \xi)$
- Reparametrization: two-parametric Fréchet

$$G_{\mathrm{Fréch}(\vartheta)}(x) = \exp\left\{ -\left(\frac{x}{\sigma}\right)^{-\alpha} \right\}$$



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Motivation and Goals

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  - Problem: disjoint block maxima make 'inefficient' use of data
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    - But why limit to maxima?
- Question: *How about using blockwise **Top**-m order statistics for GEV parameter estimation?*
    $\rightsquigarrow$ idea dates back to Weissman 1978; Smith 1986
    - Throughout literature: $(\xi_t)_{t\in\mathbb{Z}}$ assumed / approximated as i.i.d., then MLE

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- We show **lack of consistency** in general, but demonstrate a correction procedure for Top-Two order statistics

Block Maxima Modeling: Maxima

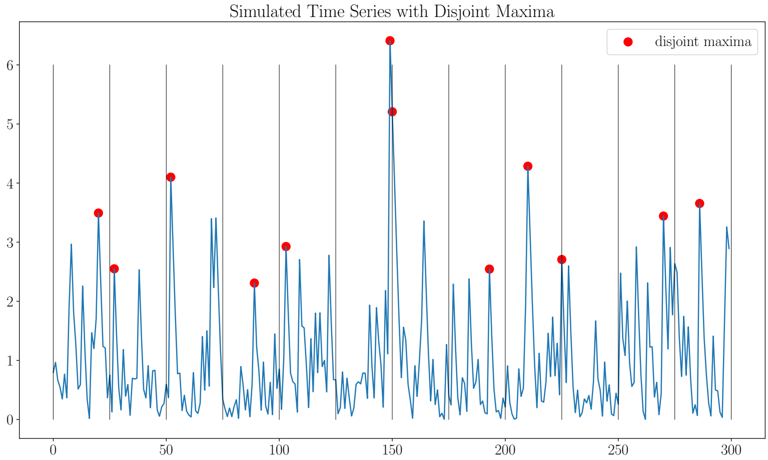

Figure 2: Sample of Disjoint Block Maxima. Distribution: GEV

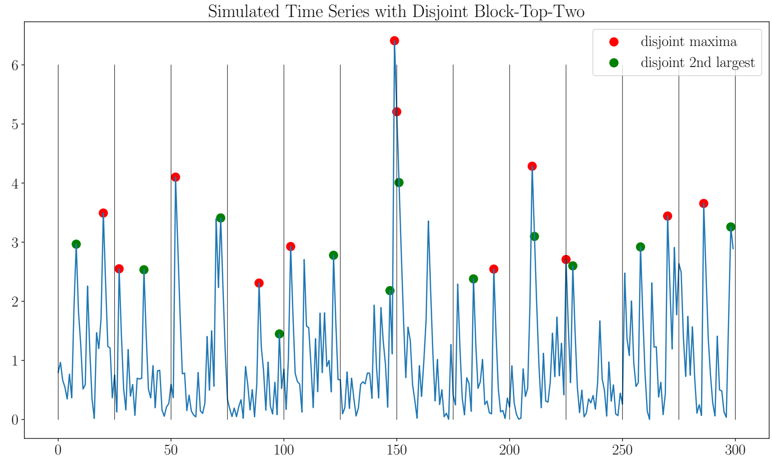

Figure 3: Sample of Disjoint Block Top-Two. Distribution: ???

Joint (limiting) Distribution of Top-Two

- Let $\xi_1, \ldots, \xi_r \subset (\xi_t)_t$, (ξ_t) stationary time series, $M_r := \xi_{r:r}$, $S_r := \xi_{r-1:r}$
- If $M_r/\sigma_r \rightsquigarrow \text{Fréchet}(\alpha, 1) =: G_\alpha(\cdot)$, how does $(M_r/\sigma_r, S_r/\sigma_r)$ behave?

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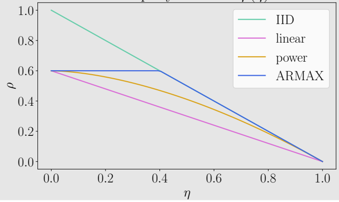
Theorem (Welsch 1972)

$$\mathbb{P}\left(M_r/\sigma_r \leq x, S_r/\sigma_r \leq y\right) \rightsquigarrow H_{\rho,\alpha,\sigma}(x,y) = \begin{cases} G_\alpha(x) & y \geq x \\ G_\alpha(y)\left\{1 - \rho(\eta_\alpha(x,y)) \log G_\alpha(y)\right\} & x > y \end{cases}$$

- $\rho : [0,1] \to [0,1]$, $\eta_\alpha(x,y) = (y/x)^\alpha$
- $\rho_0 \cdot (1-\eta) \leq \rho(\eta) \leq 1 - \eta$, *concave, non-increasing*
- ξ_t *i.i.d.* $\iff \rho(\eta) = 1 - \eta$
- *density exists* $\iff \int_0^1 \rho'(z) + z\rho''(z) \, \mathrm{d}z = -1$

Exemplary functions $\rho(\eta)$



(Pseudo-) Maximum Likelihood Estimation of Welsch-distribution

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- For some sample $(x_i, y_i)_{i=1,\ldots,n}$ compute **Pseudo**-Maximum Likelihood:

$$\ell(\alpha, \sigma) = 2k \log \alpha + 2k\alpha \log \sigma - \sum_{i=1}^{k} \left\{ (\alpha + 1) \log(x_i y_i) + \sigma^{\alpha} y_i^{-\alpha} \right\}. \tag{1}$$

  - Think of x_i: as blockmaxima of ξ_t
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  - Think of x_i: as blockmaxima of ξ_t
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- Estimator $(\hat\alpha_n, \hat\sigma_n) := \arg\max_{\alpha,\sigma} \ell(\alpha,\sigma)$, similar to Weissman 1978; Smith 1986
- Valid if assumptions violated?

(S1) Sampling scheme

- Sample 'from the limit'
- Assume $(x_i, y_i)_{i=1,\dots,k} \sim H_{\rho,\alpha_0,\sigma_n}$
  - Think of x_i: as blockmaxima of ξ_t
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$\rightsquigarrow$ Goal: high-level results for limiting distribution

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(S2) Sampling scheme

- Sample from stationary time series $\xi_1, \dots, \xi_n \subset (\xi_t)_t$
- Divide into $k = k_n$ blocks of size $r = r_n$
- Let (x_i, y_i) be top-two order statistics of block i

$\rightsquigarrow$ Goal: results on sample in DOA of $H_{\rho,\alpha_0,\sigma_n}$

- Plan: Conclude from **(S1)**

Lemma (Pseudo-MLE exists uniquely)

*The MLE $(\hat{\alpha}_k, \hat{\sigma}_k)$ of scheme **(S1)** exists uniquely with $\hat{\alpha}_k$ as root of*

$$\Psi_k(\alpha) = 2\alpha^{-1} + 2 \cdot \Big(\frac{1}{k}\sum_{i=1}^{k} y_i^{-\alpha}\Big)^{-1} \cdot \frac{1}{k}\sum_{i=1}^{k} y_i^{-\alpha} \log y_i - \frac{1}{k}\sum_{i=1}^{k} \log x_i - \frac{1}{k}\sum_{i=1}^{k} \log y_i$$

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- Weak limit of $(\hat{\alpha}_k, \hat{\sigma}_k)$ as $k \to \infty$?

Asymptotics of $(\hat{\alpha}_k, \hat{\sigma}_k)$

- Consistency? i.e. $(\hat{\alpha}_k, \hat{\sigma}_k/\sigma_k) \rightsquigarrow (\alpha_0, 1)$?
- As $k \to \infty$: $\Psi_k(\alpha) \rightsquigarrow \Psi_{\rho,\alpha_0}(\alpha)$ for some limit function Ψ depending on $\rho_0 := \rho(0)$ and α_0
- Then $\hat{\alpha}_k = \mathrm{Root}(\Psi_k(\alpha)) \rightsquigarrow \mathrm{Root}(\Psi_{\rho,\alpha_0}(\alpha)) := \alpha_1$.

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In this situation, with "$\neq$" holding unless $\rho_0 \in \{0, 1\}$,

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- Is there a way to make the estimator consistent?

Consistency correction

- Can be shown: decomposition of α_1 into $\alpha_1 = \alpha_0 \cdot \varpi(\rho_0)$
- $\varpi(\rho_0)$ only depending on ρ_0, implicitly defined as a root
- **Consistent Oracle: $\tilde{\alpha}_k := \hat{\alpha}_k / \varpi$**

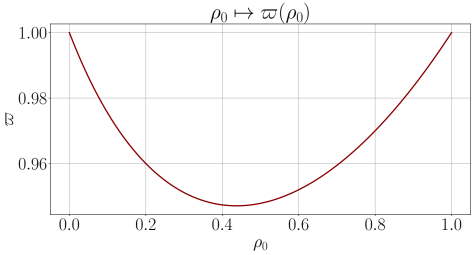

Figure 4: $\varpi : [0,1] \to [0,1]$ as function of ρ_0

Consistency correction

- **Oracle** $\tilde{\alpha}_k := \hat{\alpha}_k / \varpi$ depends on unknown $\varpi = \varpi(\rho_0)$
- How to estimate $\hat{\varpi}_k = \varpi(\hat{\rho}_{0k})$?
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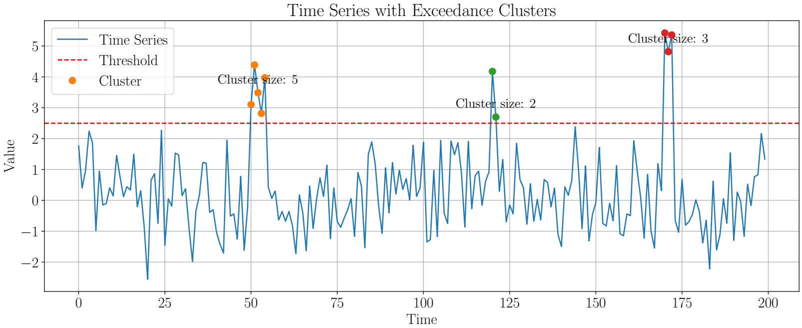
- Some estimators for $\pi(1) = \rho_0$, available; we choose $\hat{\rho}_{0k}$ by Bücher and Jennessen 2022
- modified σ_k estimator: $\tilde{\sigma}_k := \hat{\sigma}_k / \sigma_1(\hat{\rho}_{0k}, \alpha_0)$



Asymptotic distribution

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The modified estimator is consistent, i.e. $\begin{pmatrix} \tilde{\alpha}_n \\ \tilde{\sigma}_n/\sigma_n \end{pmatrix} \rightsquigarrow \begin{pmatrix} \alpha_0 \\ 1 \end{pmatrix}.$

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Theorem (General asymptotic distribution)

As $k \to \infty$, *we have under regularity conditions,*

$$\sqrt{k}\begin{pmatrix} \tilde{\alpha}_k - \alpha_0 \\ \tilde{\sigma}_k/\sigma_k - 1 \end{pmatrix} \rightsquigarrow \mathcal{N}_2(\boldsymbol{B}, \Sigma)$$

for some bias $\boldsymbol{B}$ *and covariance* Σ

(Pseudo-) MLE of Block Top-Two

(S2) Sampling Scheme

- Apply results to sample of block maxima in Fréchet-DOA
  - Sample size $k_n \to \infty$ for asymptotics
  - Block length $r_n \to \infty$ for DOA condition
  - total time series length: $n = r_n k_n$ for disjoint blocks

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  - total time series length: $n = r_n k_n$ for disjoint blocks
- sample of interest: $\boldsymbol{Z}_n = \left(\begin{pmatrix} X_{n,1} \\ Y_{n,1} \end{pmatrix} \quad \cdots \quad \begin{pmatrix} X_{n,k_n} \\ Y_{n,k_n} \end{pmatrix} \right) \overset{\text{asym}}{\sim} H_{\rho,\alpha_0,\sigma_n}$
- Think of $\boldsymbol{Z}_n$ as sample of disjoint or sliding block top-two

Block Maxima Modeling: Top-Two

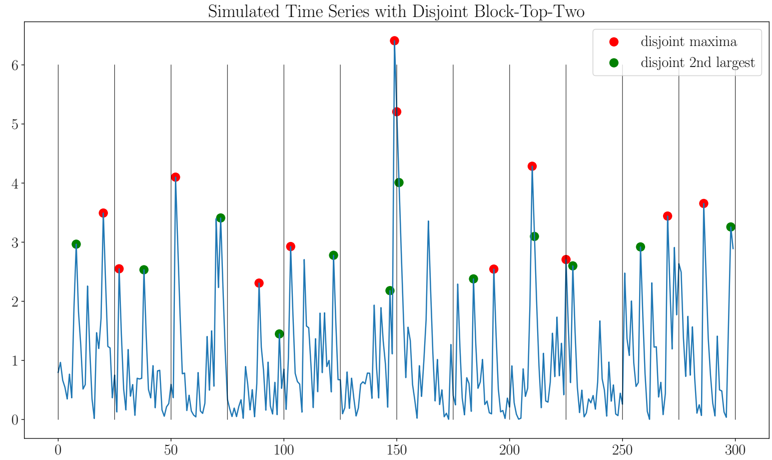

Figure 5: Sample of Disjoint Block Top-Two.

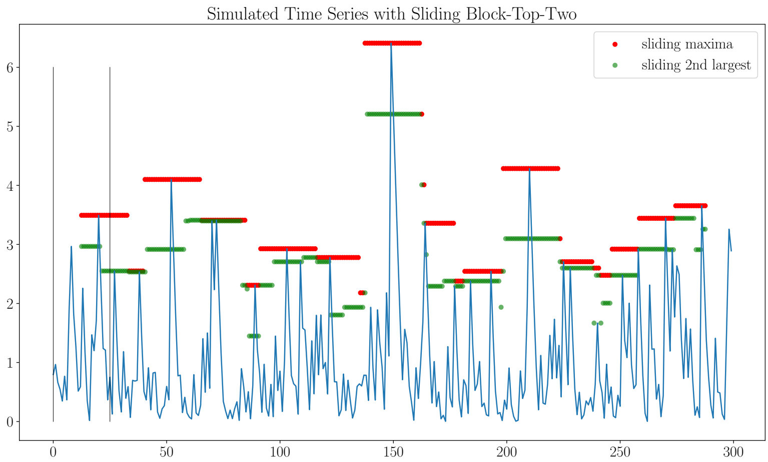

Figure 6: Sample of Sliding Block Top-Two

Asymptotic distribution: Block Maxima

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Theorem (Asymptotic Normality of Block Maxima Estimators)

Under suitable assumptions (α-mixing, finite bias, regular variation of σ_r, ...)

$$\sqrt{n/r_n}\begin{pmatrix}\tilde{\alpha}_n^{(\mathrm{mb})} - \alpha_0 \\ \tilde{\sigma}_n^{(\mathrm{mb})}/\sigma_{r_n} - 1\end{pmatrix} \rightsquigarrow \mathcal{N}_2(\boldsymbol{B}, \Sigma_{\rho,\alpha_0}^{(\mathrm{mb})})$$

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Explicit terms for i.i.d. special case?

Special Scenario: Blockwise Top-Two of I.I.D Timeseries

Special Case: IID

- Recall DOA condition $\lim_{r\to\infty} F^r(a_r x) = \exp\big(-x^{-\alpha_0}\big)$
- Equivalently: $-\log F$ is **regular varying** $\mathrm{RV}(-\alpha_0)$, i.e.

$$\lim_{u\to\infty} \frac{-\log F(ux)}{-\log F(u)} = x^{-\alpha_0}$$

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Condition [2nd Order Regular Variation]

$$\lim_{u\to\infty} \frac{1}{A(u)}\left(\frac{-\log F(ux)}{-\log F(u)} - x^{-\alpha_0}\right) = x^{-\alpha_0}\int_1^x y^{\tau-1}\,\mathrm{d}y,$$

where $A:(0,\infty)\to\mathbb{R}$, $\lim_{u\to\infty} A(u) = 0$ captures the **speed of convergence**.

Special Case: I.I.D. (Disjoint Blocks)

Theorem (Asymptotic Normality for (2nd order τ-r.v.) I.I.D. time series)

$$\sqrt{k_n}\begin{pmatrix}\tilde{\alpha}_n - \alpha_0 \\ \tilde{\sigma}_n/\sigma_n - 1\end{pmatrix} \rightsquigarrow \mathcal{N}_2\Big(M(\alpha_0)B(\alpha_0,\overline{\tau}), \Sigma_{\mathrm{DB}}^{\mathrm{TT}}\Big), \qquad n \to \infty,$$

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$$\sqrt{k_n}\begin{pmatrix}\tilde{\alpha}_n-\alpha_0\\ \tilde{\sigma}_n/\sigma_n-1\end{pmatrix}\rightsquigarrow\mathcal{N}_2\big(M(\alpha_0)B(\alpha_0,\overline{\tau}),\Sigma_{\mathrm{DB}}^{\mathrm{TT}}\big),\qquad n\to\infty,$$

with $M(\alpha_0)=\dfrac{6}{2\pi^2-3}\begin{pmatrix}\alpha_0^2 & \frac{3-2\gamma}{2}\alpha_0 & -\alpha_0^2 & -\alpha_0^2\\ \frac{2\gamma-3}{2} & \frac{3-2\pi^2-3(3-2\gamma)^2}{12\alpha_0} & \frac{3-2\gamma}{2} & \frac{3-2\gamma}{2}\end{pmatrix}$, $\overline{\tau}=\tau/\alpha_0$,

$$\Sigma_{\mathrm{DB}}^{\mathrm{TT}}\approx\begin{pmatrix}0.358\alpha_0^2 & -0.331\\ -0.331 & 0.805/\alpha_0^2\end{pmatrix},\qquad B(\alpha_0,\tau)=\frac{\lambda}{\tau\alpha_0^2}\begin{pmatrix}2\gamma-5+\Gamma(3-\tau)+\Gamma'(3-\tau)\\ \alpha_0\big(2-\Gamma(3-\tau)\big)\\ \Gamma(2-\tau)-1\\ \Gamma(1-\tau)-1\end{pmatrix}.$$

Theorem (Asymptotic Normality for (2nd order τ-r.v.) I.I.D. time series)

$$\sqrt{k_n}\begin{pmatrix}\tilde{\alpha}_n-\alpha_0\\ \tilde{\sigma}_n/\sigma_n-1\end{pmatrix}\rightsquigarrow\mathcal{N}_2\Big(M(\alpha_0)B(\alpha_0,\overline{\tau}),\boldsymbol{\Sigma}_{\mathrm{SB}}^{\mathrm{TT}}\Big),\qquad n\to\infty,$$

with $M(\alpha_0)=\dfrac{6}{2\pi^2-3}\begin{pmatrix}\alpha_0^2 & \frac{3-2\gamma}{2}\alpha_0 & -\alpha_0^2 & -\alpha_0^2\\ \frac{2\gamma-3}{2} & \frac{3-2\pi^2-3(3-2\gamma)^2}{12\alpha_0} & \frac{3-2\gamma}{2} & \frac{3-2\gamma}{2}\end{pmatrix}$, $\overline{\tau}=\tau/\alpha_0$,

$$\boldsymbol{\Sigma}_{\mathrm{SB}}^{\mathrm{TT}}\approx\begin{pmatrix}\mathbf{0.304}\boldsymbol{\alpha_0^2} & \mathbf{-0.338}\\ \mathbf{-0.338} & \mathbf{0.774}\boldsymbol{\alpha_0^2}\end{pmatrix},\qquad B(\alpha_0,\tau)=\frac{\lambda}{\tau\alpha_0^2}\begin{pmatrix}2\gamma-5+\Gamma(3-\tau)+\Gamma'(3-\tau)\\ \alpha_0\big(2-\Gamma(3-\tau)\big)\\ \Gamma(2-\tau)-1\\ \Gamma(1-\tau)-1\end{pmatrix}.$$

Comparison of asymptotic covariances

- $\mathrm{TT} \triangleq$ top-two, $\max \triangleq$ just maximum
- $\Sigma_{\mathrm{db}}^{\max}$ and $\Sigma_{\mathrm{sb}}^{\max}$ by Bücher and Segers 2018b and Bücher and Segers 2018a

$$
\begin{aligned}
\Sigma_{\mathrm{sb}}^{\mathrm{TT}} \approx \begin{pmatrix} 0.304\alpha_0^2 & -0.338 \\ -0.338 & 0.774\alpha_0^2 \end{pmatrix} <_{\mathbb{L}} \Sigma_{\mathrm{db}}^{\mathrm{TT}} &\approx \begin{pmatrix} 0.358\alpha_0^2 & -0.331 \\ -0.331 & 0.805/\alpha_0^2 \end{pmatrix} \\
<_{\mathbb{L}} \Sigma_{\mathrm{sb}}^{\max} &\approx \begin{pmatrix} 0.495\alpha_0^2 & -0.324 \\ -0.324 & 1.958/\alpha_0^2 \end{pmatrix} \\
<_{\mathbb{L}} \Sigma_{\mathrm{db}}^{\max} &\approx \begin{pmatrix} 0.608\alpha_0^2 & -0.257 \\ -0.257 & 1.109/\alpha_0^2 \end{pmatrix}
\end{aligned}
$$

- $<_{\mathbb{L}}$ denotes the LOEWNER-ordering

Simulation Results

Simulation: Shape Estimation

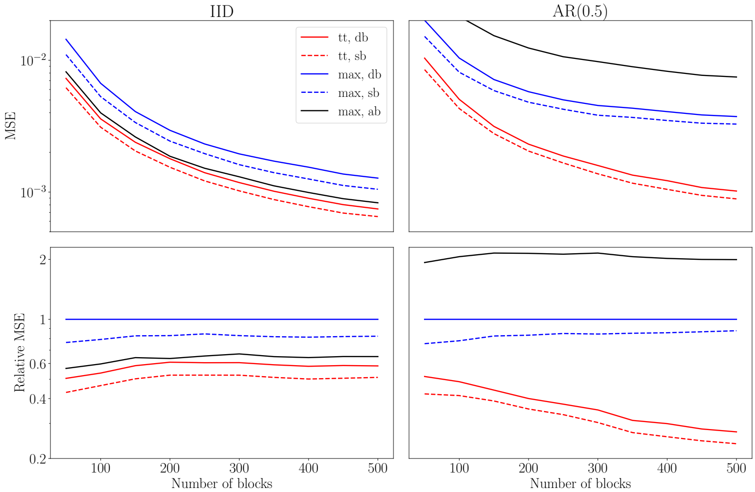

Figure 7: MSE and relative MSE, shape parameter α estimation.

Comparing our Top-Two (**tt mb**) estimators against Bücher and Segers 2018b (**max db**), Bücher and Segers 2018a (**max sb**) and Oorschot and Zhou 2020 (**max ab**)

Takeaway: **tt sb** is best

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Thank you!

Appendix

Extending to Top-r Order Statistics

Top-Two (x_i, y_i):

$$\Psi_k(\alpha) = 2\alpha^{-1} + 2 \cdot \left(\frac{1}{k}\sum_{i=1}^{k} y_i^{-\alpha}\right)^{-1} \cdot \frac{1}{k}\sum_{i=1}^{k} y_i^{-\alpha}\log y_i - \frac{1}{k}\sum_{i=1}^{k}\log x_i - \frac{1}{k}\sum_{i=1}^{k}\log y_i$$

Top-Three (x_i, y_i, z_i):

$$\Psi_k(\alpha) = 2\alpha^{-1} + 2 \cdot \left(\frac{1}{k}\sum_{i=1}^{k} z_i^{-\alpha}\right)^{-1} \cdot \frac{1}{k}\sum_{i=1}^{k} z_i^{-\alpha}\log z_i - \frac{1}{k}\sum_{i=1}^{k}\log x_i - \frac{1}{k}\sum_{i=1}^{k}\log y_i - \frac{1}{k}\sum_{i=1}^{k}\log z_i$$

Extending to Top-r Order Statistics

Top-Two (x_i, y_i):

$$\Psi_k(\alpha) = 2\alpha^{-1} + 2 \cdot \Big(\frac{1}{k}\sum_{i=1}^{k} y_i^{-\alpha}\Big)^{-1} \cdot \frac{1}{k}\sum_{i=1}^{k} y_i^{-\alpha}\log y_i - \frac{1}{k}\sum_{i=1}^{k}\log x_i - \frac{1}{k}\sum_{i=1}^{k}\log y_i$$

Top-Three (x_i, y_i, z_i):

$$\Psi_k(\alpha) = 2\alpha^{-1} + 2 \cdot \Big(\frac{1}{k}\sum_{i=1}^{k} z_i^{-\alpha}\Big)^{-1} \cdot \frac{1}{k}\sum_{i=1}^{k} z_i^{-\alpha}\log z_i - \frac{1}{k}\sum_{i=1}^{k}\log x_i - \frac{1}{k}\sum_{i=1}^{k}\log y_i - \frac{1}{k}\sum_{i=1}^{k}\log z_i$$

- Needs moments of Top-r values
- Calculation not only involves $\rho_0 = \pi(1)$, but $\pi(1), \ldots, \pi(r-1)$.

Extending to Top-r Order Statistics

Top-Two (x_i, y_i):

$$\Psi_k(\alpha) = 2\alpha^{-1} + 2 \cdot \Big(\frac{1}{k}\sum_{i=1}^{k} y_i^{-\alpha}\Big)^{-1} \cdot \frac{1}{k}\sum_{i=1}^{k} y_i^{-\alpha} \log y_i - \frac{1}{k}\sum_{i=1}^{k} \log x_i - \frac{1}{k}\sum_{i=1}^{k} \log y_i$$

Top-Three (x_i, y_i, z_i):

$$\Psi_k(\alpha) = 2\alpha^{-1} + 2 \cdot \Big(\frac{1}{k}\sum_{i=1}^{k} z_i^{-\alpha}\Big)^{-1} \cdot \frac{1}{k}\sum_{i=1}^{k} z_i^{-\alpha} \log z_i - \frac{1}{k}\sum_{i=1}^{k} \log x_i - \frac{1}{k}\sum_{i=1}^{k} \log y_i - \frac{1}{k}\sum_{i=1}^{k} \log z_i$$

- Needs moments of Top-r values
- Calculation not only involves $\rho_0 = \pi(1)$, but $\pi(1), \ldots, \pi(r-1)$.
- Rest should work in analogy
- Outlook: **How to choose r in practice?**

Simulation: Scale Estimation

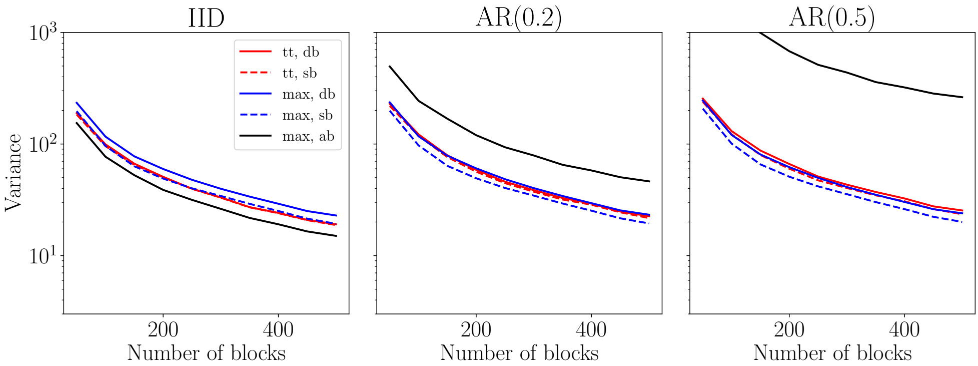

Figure 8: MSE under increasing temporal dependence, scale parameter σ estimation.

Takeaway: max sb is best

- Often interested in estimating (e.g.) [100-year]-return levels $\mathrm{RL}(100)$
  - the value, which is exceeded once every 100 years (in expectation)

$$
\begin{aligned}
\mathrm{RL}(T) := F^{\leftarrow}(1-1/T) &= \inf\{x \in \mathbb{R} : F(x) \geq 1-1/T\} \\
&= \sigma(-\log(1-1/T))^{-1/\alpha}
\end{aligned}
$$

- being function of (α, σ): $\mathrm{RL}(T) = \mathrm{RL}(T)(\alpha, \sigma) \rightsquigarrow$ need estimation of both parameters

- Often interested in estimating (e.g.) [100-year]-return levels $\mathrm{RL}(100)$
  - the value, which is exceeded once every 100 years (in expectation)

$$\begin{aligned} \mathrm{RL}(T) &:= F^{\leftarrow}(1 - 1/T) = \inf\{x \in \mathbb{R} : F(x) \geq 1 - 1/T\} \\ &= \sigma(-\log(1 - 1/T))^{-1/\alpha} \end{aligned}$$

- being function of (α, σ): $\mathrm{RL}(T) = \mathrm{RL}(T)(\alpha, \sigma) \rightsquigarrow$ need estimation of both parameters
- Estimate α via **tt mb** and σ via **max sb**
  - $\widehat{\mathrm{RL}}(T) = \mathrm{RL}(T)(\hat{\alpha}_{\textbf{tt mb}}, \hat{\sigma}_{\textbf{max sb}})$
- Does the resulting estimator combine 'best of two worlds'? **BOTWE**?

Simulation: Return Level Estimation

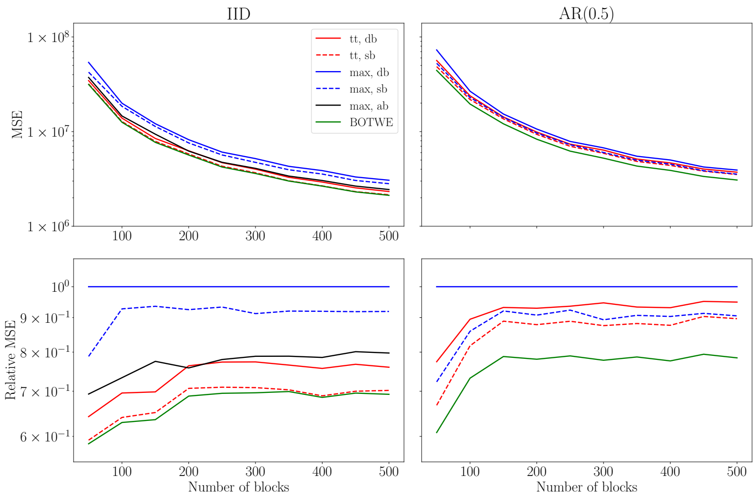

Figure 9: MSE and relative MSE, return level $\mathrm{RL}(100)$ estimation.

BOTWE: estimate α via **tt sb** and σ_r via **max sb** ('best of two worlds')

Takeaway:

BOTWE is indeed **BOTWE**!

Appendix: Fixed-n simulation study

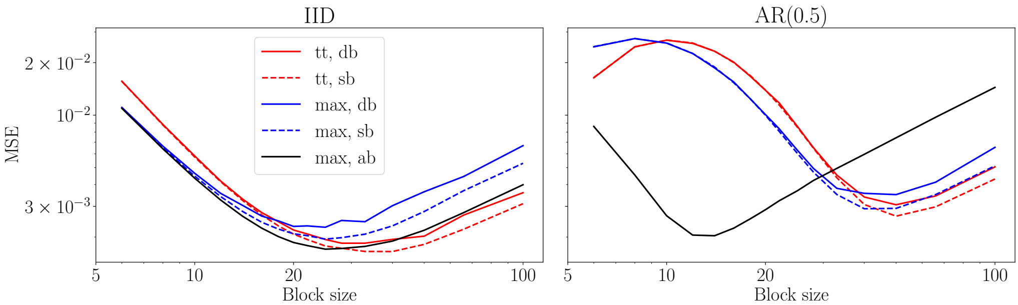

Figure 10: MSE for growing block size r and fixed total sample size n